

Description:

- Dataset: final_dataset_rate.csv
- 재무비율 및 기업 정보 관련 변수만 들어 있는, "전처리된" dataset. (NA 채우기 등.)
- Target: Bad (파산 or 부도), bad event가 있었던 기업이라도 event가 일어나기 전 해에는 0으로 처리.
- Duration: 각각의 기업에 대해, 회계년 - 최초기록연도 (for 생존분석)

```
In [1]: import pandas as pd
import numpy as np
import random
from sklearn.preprocessing import StandardScaler

from sklearn.utils import resample
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import StratifiedKFold, train_test_split
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, roc_auc_score, f1_score

import copy
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import DataLoader, TensorDataset
from sklearn.model_selection import TimeSeriesSplit
```

1. Dataset Preview & 추가 전처리

```
In [2]: df = pd.read_csv('final_dataset_rate.csv')

print(df.shape)
print(df.head())
```

(96161, 24)

	Symbol	Name	결산월	회계년	상장폐지일자	
폐지사유 \						
0	A000010	조흥은행	12.0	1999	2004-07-02T00:00:00Z	지주회사의 완전자회사화 (지주회사 신규상장)
1	A000010	조흥은행	12.0	2000	2004-07-02T00:00:00Z	지주회사의 완전자회사화 (지주회사 신규상장)
2	A000010	조흥은행	12.0	2001	2004-07-02T00:00:00Z	지주회사의 완전자회사화 (지주회사 신규상장)
3	A000010	조흥은행	12.0	2002	2004-07-02T00:00:00Z	지주회사의 완전자회사화 (지주회사 신규상장)
4	A000010	조흥은행	12.0	2003	2004-07-02T00:00:00Z	지주회사의 완전자회사화 (지주회사 신규상장)

	파산여부	부도여부	Bad	duration	...	영업이익/총자본	유보이익/총자본	자	
기자본비율		자본금회전율							
0	0	0	0	0	...	-0.417365	-0.469008	0.046722	1.062795
1	0	0	0	1	...	0.078942	-0.387289	0.041104	1.151435
2	0	0	0	2	...	0.310515	-0.133537	0.045285	1.060018
3	0	0	0	3	...	-0.044457	-0.380895	0.034496	1.010100
4	0	0	0	4	...	-0.348575	-0.945519	0.032365	0.972309

	차입금의존도	총부채회전율	총자본투자효율	총자본회전율	총순자산수익률	
현금성자산비율						
0	0.0	0.080646	-0.019500	0.076878	-0.337671	0.025787
1	0.0	0.079727	0.003245	0.076450	0.046268	0.019043
2	0.0	0.066594	0.014062	0.063578	0.205990	0.023778
3	0.0	0.052808	-0.001534	0.050987	-0.251902	0.017466
4	0.0	0.059618	-0.011282	0.057689	-0.490501	0.015151

[5 rows x 24 columns]

```
In [3]: print(f"columns: {df.columns.to_list()}")
print(f"상장폐지 기업 개수: {df['Bad'].sum()}") # Label 1 개수 (0, 1)
```

columns: ['Symbol', 'Name', '결산월', '회계년', '상장폐지일자', '폐지사유', '파산여부', '부도여부', 'Bad', 'duration', '금융비용/부채', '금융비용/총비용', '설비투자효율', '현금흐름/총자본', '영업이익/총자본', '유보이익/총자본', '자기자본비율', '자본금회전율', '차입금의존도', '총부채회전율', '총자본투자효율', '총자본회전율', '총순자산수익률', '현금성자산비율']

상장폐지 기업 개수: 70

- Symbol: 기업 코드
- 회계년: 연도 (Time-Series)
- 상장폐지일자: 상장폐지 이후 data 모두 삭제 필요 [회생 등 무시]
- Bad: 0 or 1 == target feature label
- '금융비용/부채' - '현금성자산비율': input features

Question. 상장폐지 이후 data 존재하는 기업?

```
In [4]: # 1. Bad == 1을 가진 기업(Symbol) List
symbols_with_bad_1 = df[df['Bad'] == 1]['Symbol'].unique()

# 2. Bad == 1 이후에도 데이터가 있는 기업들?
symbols_with_post_bad_rows = []

# 3. Symbol별로 순회하며 체크
for symbol in symbols_with_bad_1:
```

```

group = df[df['Symbol'] == symbol] # 해당 Symbol의 모든 row
bad_idx = group[group['Bad'] == 1].index

if not bad_idx.empty:
    first_bad_idx = bad_idx[0]
    # Bad == 1 이후에도 행이 존재하는 경우
    if first_bad_idx < group.index.max():
        symbols_with_post_bad_rows.append(symbol)

# 4. 결과 확인
print("Bad == 1 이후에도 데이터가 있는 기업 목록:")
print(symbols_with_post_bad_rows)

```

Bad == 1 이후에도 데이터가 있는 기업 목록:
 []

- 상장폐지 이후 data 모두 삭제된 상태이므로, NA 체크

```

In [5]: na_count = df.isna().sum()
na_ratio = df.isna().mean()

na_summary = pd.DataFrame({
    '결측값 개수': na_count,
    '결측값 비율': na_ratio
})

na_summary['결측값 비율'] = (na_summary['결측값 비율'] * 100).round(2).astype(str)
print(na_summary)

```

	결측값 개수	결측값 비율
Symbol	0	0.0%
Name	0	0.0%
결산월	29861	31.05%
회계년	0	0.0%
상장폐지일자	83319	86.65%
폐지사유	83521	86.86%
파산여부	0	0.0%
부도여부	0	0.0%
Bad	0	0.0%
duration	0	0.0%
금융비용/부채	0	0.0%
금융비용/총비용	0	0.0%
설비투자효율	0	0.0%
현금흐름/총자본	0	0.0%
영업이익/총자본	0	0.0%
유보이익/총자본	0	0.0%
자기자본비율	0	0.0%
자본금회전율	0	0.0%
차입금의존도	0	0.0%
총부채회전율	0	0.0%
총자본투자효율	0	0.0%
총자본회전율	0	0.0%
총순자산수익률	0	0.0%
현금성자산비율	0	0.0%

- 결산월의 경우 사용되는 feature가 아니라서 무시
- 폐지사유의 경우에도 이미 Bad를 target으로 정의할 때 사용했으니 무시

- 상장폐지일자는 뭘까.... 체크
 - 1. 비상장폐지 기업의 경우 모든 연도 NA
 - 2. 상장폐지 기업의 경우 모든 연도에서 같은 일자

```
In [6]: # 상장폐지한 기업
delisted_symbols = df[df['상장폐지일자'].notna()]['Symbol'].unique()

# 상장폐지 기업 수
n_delisted = len(delisted_symbols)

# 해당 기업들이 가진 row 수
n_delisted_rows = df[df['Symbol'].isin(delisted_symbols)].shape[0]

print(f"(1) 상장폐지 기업 수: {n_delisted}")
print(f"(1) 상장폐지 기업이 차지하는 row 수: {n_delisted_rows}\n")

#-----#

# Bad == 1인 기업
bad_1_symbols = df[df['Bad'] == 1]['Symbol'].unique()

# Bad == 1 기업 수
n_bad_1 = len(bad_1_symbols)

# 해당 기업들이 가진 row 수
n_bad_1_rows = df[df['Symbol'].isin(bad_1_symbols)].shape[0]

print(f"(2) Bad=1 기업 수: {n_bad_1}")
print(f"(2) Bad=1 기업이 차지하는 row 수: {n_bad_1_rows}\n")

#-----#

# 전체 기업 수
n_total_symbols = df['Symbol'].nunique()

# 전체 row 수
n_total_rows = df.shape[0]

print(f"(3) 전체 기업 수: {n_total_symbols}")
print(f"(3) 전체 row 수: {n_total_rows}\n")

#-----#

print(f"(4) 상장폐지 row 비율: {n_delisted_rows / n_total_rows:.2%}")
```

- (1) 상장폐지 기업 수: 1342
 (1) 상장폐지 기업이 차지하는 row 수: 12842
- (2) Bad=1 기업 수: 70
 (2) Bad=1 기업이 차지하는 row 수: 574
- (3) 전체 기업 수: 5603
 (3) 전체 row 수: 96161
- (4) 상장폐지 row 비율: 13.35%

- (4)를 보니, 상장폐지일자에 있는 NA는 정당함!. 상장폐지했는데 NA로 기록된 사례는 없음.

Features 정제

- 상장폐지일자 -> 상장폐지년도로 변경.
- 회계년과 같은 연도단위로 사용하기 위함
- 안쓰는 feature 삭제
- ratio feature -> 표준화

```
In [7]: # 상장폐지일자 -> 상장폐지년도로 변경.
df["상장폐지년도"] = pd.to_datetime(df["상장폐지일자"], errors="coerce").dt.year

drop_columns = ['Name', '결산월', '상장폐지일자', '폐지사유', '파산여부', '부도여부']
df = df.drop(columns=drop_columns)

print(f"columns: {df.columns.to_list()}")
```

columns: ['Symbol', '회계년', 'Bad', 'duration', '금융비용/부채', '금융비용/총비용', '설비투자효율', '현금흐름/총자본', '영업이익/총자본', '유보이익/총자본', '자기자본비율', '자본금회전율', '차입금의존도', '총부채회전율', '총자본투자효율', '총자본회전율', '총순자산수익률', '현금성자산비율', '상장폐지년도']

```
In [8]: columns = df.columns.to_list()

# columns 순서 이쁘게 정렬
columns = columns[:2] + [ columns[-1] ] + columns[3:-1] + [ columns[2] ]

print(f"columns: {columns}")
df = df[columns]
```

columns: ['Symbol', '회계년', '상장폐지년도', 'duration', '금융비용/부채', '금융비용/총비용', '설비투자효율', '현금흐름/총자본', '영업이익/총자본', '유보이익/총자본', '자기자본비율', '자본금회전율', '차입금의존도', '총부채회전율', '총자본투자효율', '총자본회전율', '총순자산수익률', '현금성자산비율', 'Bad']

```
In [9]: scaled_columns = columns[4:-1]
scaler = StandardScaler()
df_scaled_values = scaler.fit_transform(df[scaled_columns])

df_scaled = df.copy()
df_scaled[scaled_columns] = df_scaled_values
```

- 새로운 dataset file로 저장 (안하셔도 됩니다)
 - Columns Sorted
 - Columns Selected
 - Standardized Ratio Features

```
In [10]: # csv로 저장
df.to_csv("managed.csv", index=False, encoding='utf-8-sig')
```

2. Non-Time Series Analysis

⇒ label의 경우 Bad event 이전은 모두 0으로 처리

- e.g. (Bad event 발생) Label = 0, 0, 0, ..., 1
- e.g. (Bad event 발생 X) Label = 0, 0, 0, ..., 0

Label Ratio (Case Sampling)

```
In [10]: input_features = columns[:-1]
target_feature = columns[-1]

print(f"input features: {input_features}")
print(f"target feature: {target_feature}")

# label distribution
counts = df[target_feature].value_counts(normalize=False).sort_index()
ratios = df[target_feature].value_counts(normalize=True).sort_index()

for label in counts.index:
    count = counts[label]
    ratio = ratios[label]
    print(f'Label {label}: {count} 개 ({ratio:.2%})')
```

input features: ['Symbol', '회계년', '상장폐지년도', 'duration', '금융비용/부채', '금융비용/총비용', '설비투자효율', '현금흐름/총자본', '영업이익/총자본', '유보이익/총자본', '자기자본비율', '자본금회전율', '차입금의존도', '총부채회전율', '총자본회전율', '총순자산수익률', '현금성자산비율']

target feature: Bad

Label 0: 96091 개 (99.93%)

Label 1: 70 개 (0.07%)

```
In [11]: print(df.head())
```

	Symbol	회계년	상장폐지년도	duration	금융비용/부채	금융비용/총비용	설 비투자효율	현금흐름/총자본	\
0	A000010	1999	2004	0	0.0	0.0	2.493567	0.112994	
1	A000010	2000	2004	1	0.0	0.0	2.853872	0.720505	
2	A000010	2001	2004	2	0.0	0.0	2.668069	0.643380	
3	A000010	2002	2004	3	0.0	0.0	2.619010	0.704620	
4	A000010	2003	2004	4	0.0	0.0	2.715868	0.903742	

	영업이익/총자본	유보이익/총자본	자기자본비율	자본금회전율	차입금의존도	총부채회전율	총자본투자효율	\
0	-0.417365	-0.469008	0.046722	1.062795	0.0	0.080646	-0.019500	
1	0.078942	-0.387289	0.041104	1.151435	0.0	0.079727	0.003245	
2	0.310515	-0.133537	0.045285	1.060018	0.0	0.066594	0.014062	
3	-0.044457	-0.380895	0.034496	1.010100	0.0	0.052808	-0.001534	
4	-0.348575	-0.945519	0.032365	0.972309	0.0	0.059618	-0.011282	

	총자본회전율	총순자산수익률	현금성자산비율	Bad
0	0.076878	-0.337671	0.025787	0
1	0.076450	0.046268	0.019043	0
2	0.063578	0.205990	0.023778	0
3	0.050987	-0.251902	0.017466	0
4	0.057689	-0.490501	0.015151	0

Matched-Pairs Sampling 기법 사용 (1:k)

- Bad event 발생기업 1 : 비 발생기업 k matching

- Label 1 개수가 매우 적으므로, k = 5 사용

Symbol	회계년	상장폐지년도	Bad
A	2020	2020	1
B	2020	NaN	0
C	2020	NaN	0
D	2020	NaN	0
E	2020	NaN	0
F	2020	NaN	0

```
In [11]: # 1. Bad = 1 기업 목록
bad_firms = df[df['Bad'] == 1]['Symbol'].unique()

# 2. Bad event 발생연도 정보 (Symbol, 회계년 = 상장폐지년도)
bad_events = df[df['Bad'] == 1][['Symbol', '상장폐지년도']].copy()

# 3. Bad 비발생 기업 Symbol 추출
symbol_bad_sum = df.groupby('Symbol')['Bad'].sum()
good_firms = symbol_bad_sum[symbol_bad_sum == 0].index

# 4. Bad 비발생 기업
df_good = df[df['Symbol'].isin(good_firms)].copy()

# 5. 매칭 시작
matched_neg_list = []

for _, row in bad_events.iterrows():
    fail_year = row['상장폐지년도']
    fail_symbol = row['Symbol']

    # 해당 연도에 존재하는 정상 기업 중 5개
    candidates = df_good[df_good['회계년'] == fail_year]

    if len(candidates) >= 5:
        matched = candidates.sample(n=5, replace=False, random_state=42)
        matched['matched_to'] = fail_symbol # 어떤 Bad event 발생기업과 매칭됐는
        matched_neg_list.append(matched)

# 6. 최종 결합
df_neg_matched = pd.concat(matched_neg_list, ignore_index=True)
df_pos = df[df['Bad'] == 1].copy()
df_matched = pd.concat([df_pos, df_neg_matched], ignore_index=True)

In [13]: print(df_pos.head())
```

	Symbol	회계년	상장폐지년도	duration	금융비용/부채	금융비용/총비용
설비투자효율 \						
5812	A001150	2003	2003	4	0.301006	0.349108
6661	A001310	2012	2012	13	0.140968	0.086570
9868	A001890	2004	2004	5	0.117265	0.182017
10194	A001980	2002	2004	3	0.101917	0.119617
10726	A002130	2001	2001	2	0.069252	0.430069

	현금흐름/총자본	영업이익/총자본	유보이익/총자본	자기자본비율	자본금회전율	차입금의존도	총부채회전율 \
5812	0.507908	0.231577	1.437455	-2.806307	1.763127	1.450316	0.691477
6661	-1.639224	-6.348441	-13.294435	0.113255	6.210845	0.049293	0.817553
9868	13.084650	6.942818	44.648249	-0.011793	0.946780	0.415684	0.563331
10194	-1.285327	1.950647	-15.500868	0.032272	1.603687	0.543411	0.917078
10726	0.007613	0.096627	1.145621	-1.301494	1.270873	0.021866	0.106383

	총자본투자효율	총자본회전율	총순자산수익률	현금성자산비율	Bad
5812	-0.649876	2.631974	1.337826	0.044577	1
6661	-0.718991	0.724961	-13.171735	0.042940	1
9868	-0.081875	0.569974	24.221995	0.009644	1
10194	0.062951	0.887482	11.319867	0.065962	1
10726	-0.125760	0.244839	0.143601	0.000000	1

```
In [14]: print(df_neg_matched.head())
```

	Symbol	회계년	상장폐지년도	duration	금융비용/부채	금융비용/총비용
0	A022520	2003	2012	3	0.013185	0.003096
1	A016140	2003	2012	4	0.055002	0.091426
2	A025560	2003	<NA>	4	0.073620	0.040310
3	A050120	2003	<NA>	3	0.012008	0.003868
4	A053210	2003	<NA>	2	0.049081	0.066368

	영업이익/총자본	유보이익/총자본	자기자본비율	자본금회전율	차입금의존도	총부채회전율	총자본투자효율 \
0	0.013005	0.239398	0.578152	12.224100	0.013676	4.276232	0.007519
1	0.089248	0.309957	0.512842	5.194044	0.240555	0.695556	0.045770
2	-0.031844	0.015885	0.762103	5.304692	0.121699	1.724317	-0.024268
3	0.046711	0.447526	0.578132	8.480126	0.090747	3.168539	0.027005
4	-1.217237	-2.193916	0.241514	0.395239	0.147209	0.351945	-0.293980

	총자본회전율	총순자산수익률	현금성자산비율	Bad	matched_to
0	1.803919	0.010498	0.108289	0	A001150
1	0.338845	0.080175	0.001475	0	A001150
2	0.410210	0.005084	0.093084	0	A001150
3	1.336706	0.049254	0.019935	0	A001150
4	0.266945	-1.244288	0.343658	0	A001150

```
In [15]: print(df_matched.head())
```

	Symbol	회계년	상장폐지년도	duration	금융비용/부채	금융비용/총비용	설	
비	투자효율	현금흐름/총자본	\					
0	A001150	2003	2003	4	0.301006	0.349108	7.353275	0.507908
1	A001310	2012	2012	13	0.140968	0.086570	2.789950	-1.639224
2	A001890	2004	2004	5	0.117265	0.182017	1.437477	13.084650
3	A001980	2002	2004	3	0.101917	0.119617	4.757537	-1.285327
4	A002130	2001	2001	2	0.069252	0.430069	0.385966	0.007613

	영업이익/총자본		유보이익/총자본		자기자본비율		자본금회전율		차입금의존	
도	총부채회전율		총자본투자효율		\					
0	0.231577	1.437455	-2.806307	1.763127	1.450316	0.691477	-0.649876			
1	-6.348441	-13.294435	0.113255	6.210845	0.049293	0.817553	-0.718991			
2	6.942818	44.648249	-0.011793	0.946780	0.415684	0.563331	-0.081875			
3	1.950647	-15.500868	0.032272	1.603687	0.543411	0.917078	0.062951			
4	0.096627	1.145621	-1.301494	1.270873	0.021866	0.106383	-0.125760			

	총자본회전율		총순자산수익률	현금성자산비율	Bad matched_to
0	2.631974	1.337826	0.044577	1	NaN
1	0.724961	-13.171735	0.042940	1	NaN
2	0.569974	24.221995	0.009644	1	NaN
3	0.887482	11.319867	0.065962	1	NaN
4	0.244839	0.143601	0.000000	1	NaN

Correlation 기반 Feature Importance

```
In [12]: X = df_matched[input_features[4:]]
y = df_matched[target_feature]

correlation_matrix = X.corr() # 기본값: Pearson correlation
print(correlation_matrix)

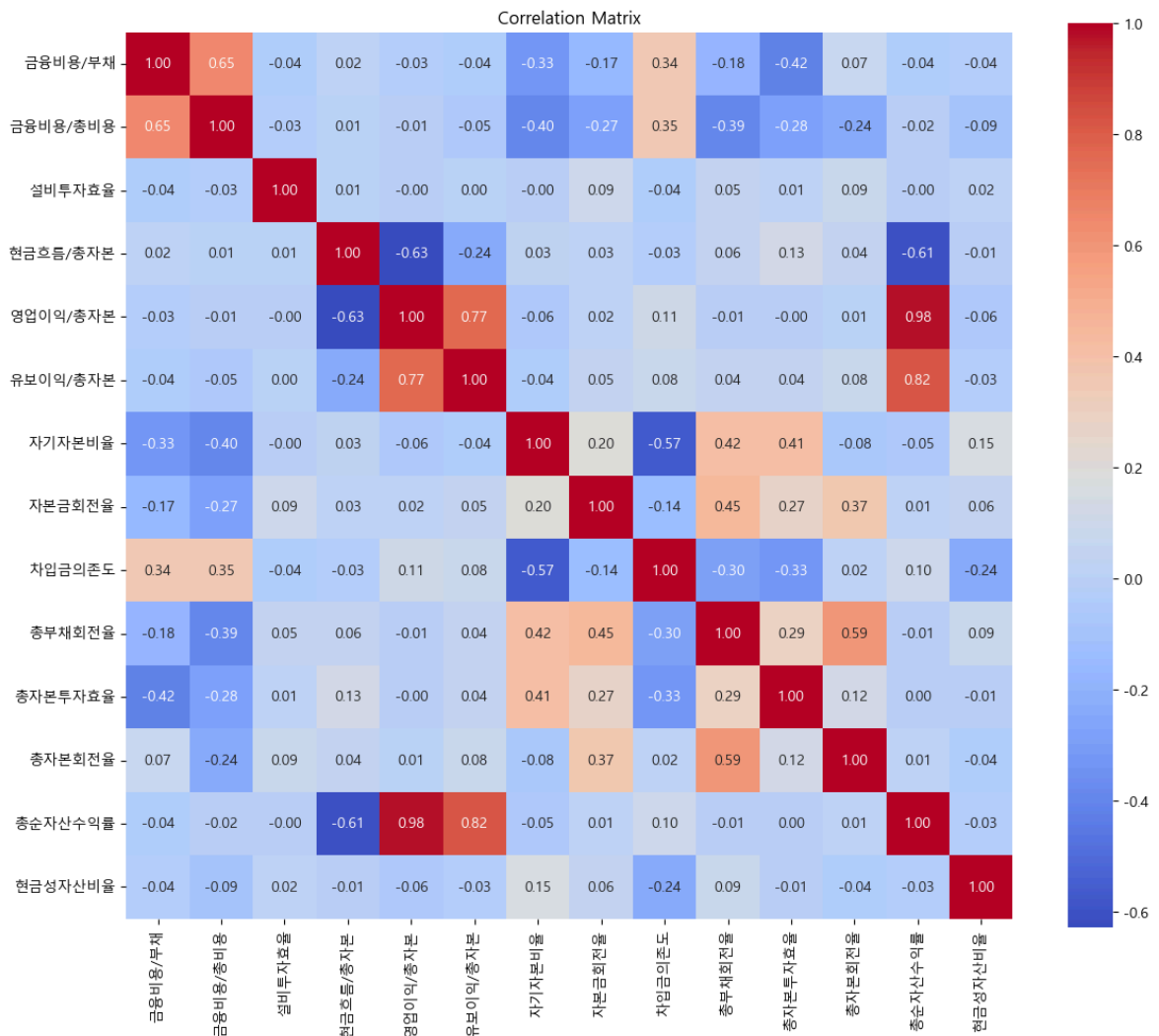
# font
plt.rcParams['font.family'] = 'Malgun Gothic'
plt.rcParams['axes.unicode_minus'] = False

# Heat map
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, fmt=".2f", cmap='coolwarm', square=True)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

	금융비용/부채	금융비용/총비용	설비투자효율	현금흐름/총자본	영업이익/총자본
유보이익/총자본	\				
금융비용/부채	1.000000	0.647416	-0.040934	0.020968	-0.034984
금융비용/총비용	0.647416	1.000000	-0.034229	0.006660	-0.010960
설비투자효율	-0.040934	-0.034229	1.000000	0.005581	-0.003391
현금흐름/총자본	0.020968	0.006660	0.005581	1.000000	-0.627755
영업이익/총자본	-0.034984	-0.010960	-0.003391	-0.627755	1.000000
유보이익/총자본	-0.042043	-0.045005	0.001986	-0.243686	0.771022
자기자본비율	-0.325486	-0.404689	-0.002225	0.027586	-0.061046
자본금회전율	-0.167371	-0.273380	0.093653	0.025416	0.019937
차입금의존도	0.341638	0.348279	-0.040054	-0.025500	0.113832
총부채회전율	-0.179799	-0.390516	0.050980	0.057807	-0.013986
총자본투자효율	-0.420451	-0.278192	0.009930	0.125824	-0.000712
총자본회전율	0.066866	-0.243426	0.088295	0.038407	0.006395
총순자산수익률	-0.041450	-0.015228	-0.002342	-0.606725	0.977675
현금성자산비율	-0.035676	-0.092285	0.019768	-0.012789	-0.058865

	자기자본비율	자본금회전율	차입금의존도	총부채회전율	총자본	
투자효율	총자본회전율	\				
금융비용/부채	-0.325486	-0.167371	0.341638	-0.179799	-0.420451	0.066866
금융비용/총비용	-0.404689	-0.273380	0.348279	-0.390516	-0.278192	-0.243426
설비투자효율	-0.002225	0.093653	-0.040054	0.050980	0.009930	0.088295
현금흐름/총자본	0.027586	0.025416	-0.025500	0.057807	0.125824	0.038407
영업이익/총자본	-0.061046	0.019937	0.113832	-0.013986	-0.000712	0.006395
유보이익/총자본	-0.039942	0.050146	0.084344	0.043041	0.041222	0.075581
자기자본비율	1.000000	0.196836	-0.565009	0.418785	0.407015	-0.075471
자본금회전율	0.196836	1.000000	-0.144432	0.449534	0.270658	0.365133
차입금의존도	-0.565009	-0.144432	1.000000	-0.296486	-0.333407	0.023419
총부채회전율	0.418785	0.449534	-0.296486	1.000000	0.289369	0.592956
총자본투자효율	0.407015	0.270658	-0.333407	0.289369	1.000000	0.124664
총자본회전율	-0.075471	0.365133	0.023419	0.592956	0.124664	1.000000
총순자산수익률	-0.053970	0.010342	0.100206	-0.011571	0.000955	0.010344
현금성자산비율	0.148976	0.057206	-0.242686	0.088937	-0.010073	-0.042660

	총순자산수익률	현금성자산비율
금융비용/부채	-0.041450	-0.035676
금융비용/총비용	-0.015228	-0.092285
설비투자효율	-0.002342	0.019768
현금흐름/총자본	-0.606725	-0.012789
영업이익/총자본	0.977675	-0.058865
유보이익/총자본	0.817821	-0.034399
자기자본비율	-0.053970	0.148976
자본금회전율	0.010342	0.057206
차입금의존도	0.100206	-0.242686
총부채회전율	-0.011571	0.088937
총자본투자효율	0.000955	-0.010073
총자본회전율	0.010344	-0.042660
총순자산수익률	1.000000	-0.033149
현금성자산비율	-0.033149	1.000000



- 다중공선성 기반 변수 소거
- Correlation 크기 > 0.8인 경우 소거

```
In [13]: corr_matrix = X.corr().abs()

upper = corr_matrix.where(np.triu(np.ones(corr_matrix.shape), k=1).astype(bool))

to_drop = [column for column in upper.columns if any(upper[column] > 0.8)]

print("reduced input features list:")
print(to_drop)

X_reduced = X.drop(columns=to_drop)

print("\ninput features change:")
print(X.shape)
print(X_reduced.shape)

print('_____')
reduced_correlation_matrix = X_reduced.corr()
print(reduced_correlation_matrix)

# font
plt.rcParams['font.family'] = 'Malgun Gothic'
plt.rcParams['axes.unicode_minus'] = False
```

```
# Heat map
plt.figure(figsize=(12, 10))
sns.heatmap(reduced_correlation_matrix, annot=True, fmt=".2f", cmap='coolwarm',
plt.title("Reduced Correlation Matrix")
plt.tight_layout()
plt.show()
```

reduced input features list:

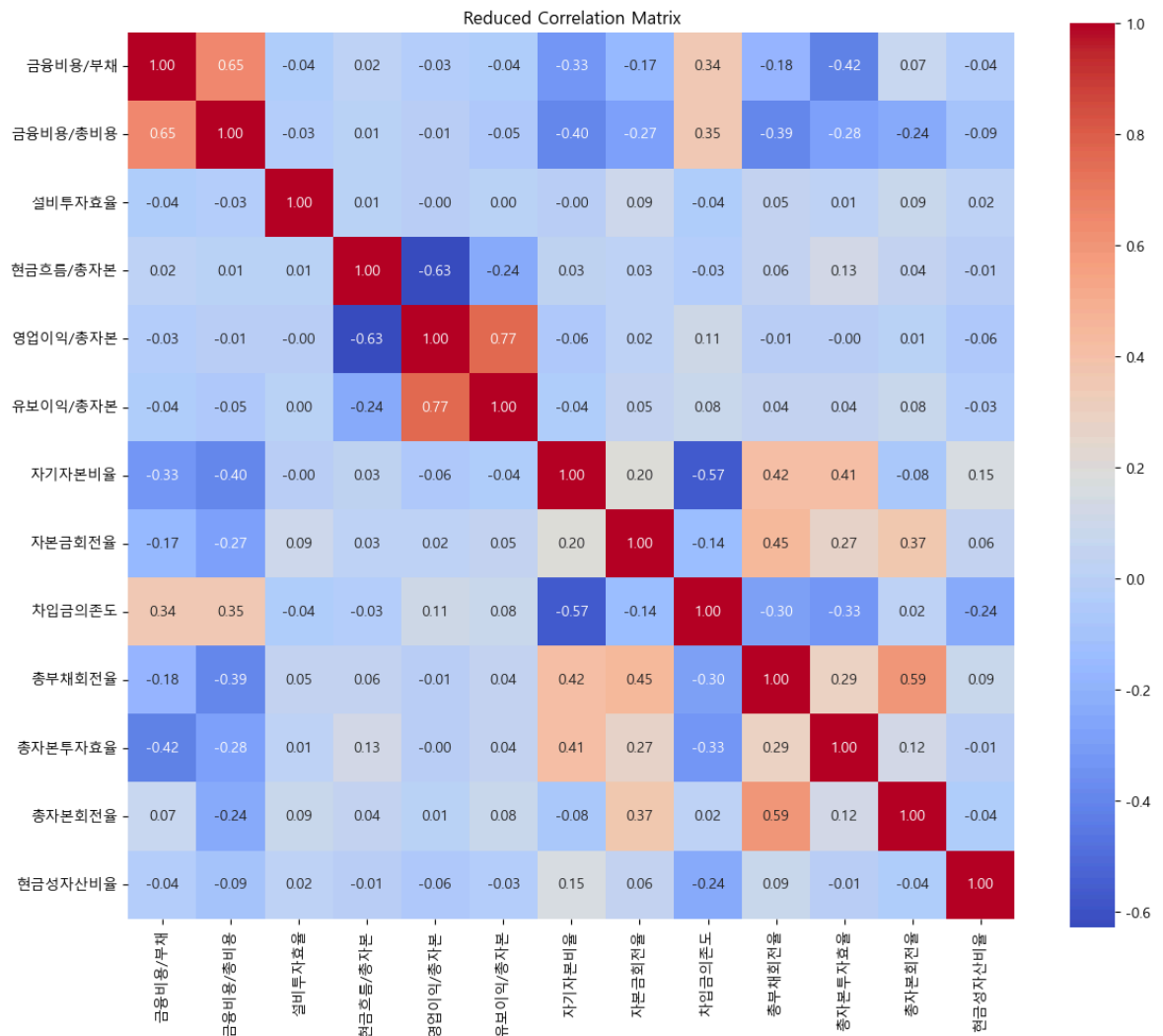
['총순자산수익률']

input features change:

(420, 14)

(420, 13)

	금융비용/부채	금융비용/총비용	설비투자효율	현금흐름/총자본	영업이익/총자본	유보이익/총자본	\
금융비용/부채	1.000000	0.647416	-0.040934	0.020968	-0.034984	-0.042043	
금융비용/총비용	0.647416	1.000000	-0.034229	0.006660	-0.010960	-0.045005	
설비투자효율	-0.040934	-0.034229	1.000000	0.005581	-0.003391	0.001986	
현금흐름/총자본	0.020968	0.006660	0.005581	1.000000	-0.627755	-0.243686	
영업이익/총자본	-0.034984	-0.010960	-0.003391	-0.627755	1.000000	0.771022	
유보이익/총자본	-0.042043	-0.045005	0.001986	-0.243686	0.771022	1.000000	
자기자본비율	-0.325486	-0.404689	-0.002225	0.027586	-0.061046	-0.039942	
자본금회전율	-0.167371	-0.273380	0.093653	0.025416	0.019937	0.050146	
차입금의존도	0.341638	0.348279	-0.040054	-0.025500	0.113832	0.084344	
총부채회전율	-0.179799	-0.390516	0.050980	0.057807	-0.013986	0.043041	
총자본투자효율	-0.420451	-0.278192	0.009930	0.125824	-0.000712	0.041222	
총자본회전율	0.066866	-0.243426	0.088295	0.038407	0.006395	0.075581	
현금성자산비율	-0.035676	-0.092285	0.019768	-0.012789	-0.058865	-0.034399	
	자기자본비율	자본금회전율	차입금의존도	총부채회전율	총자본투자효율	총자본회전율	현금성자산비율
금융비용/부채	-0.325486	-0.167371	0.341638	-0.179799	-0.420451	0.066866	-0.035676
금융비용/총비용	-0.404689	-0.273380	0.348279	-0.390516	-0.278192	-0.243426	-0.092285
설비투자효율	-0.002225	0.093653	-0.040054	0.050980	0.009930	0.088295	0.019768
현금흐름/총자본	0.027586	0.025416	-0.025500	0.057807	0.125824	0.038407	-0.012789
영업이익/총자본	-0.061046	0.019937	0.113832	-0.013986	-0.000712	0.006395	-0.058865
유보이익/총자본	-0.039942	0.050146	0.084344	0.043041	0.041222	0.075581	-0.034399
자기자본비율	1.000000	0.196836	-0.565009	0.418785	0.407015	-0.075471	0.148976
자본금회전율	0.196836	1.000000	-0.144432	0.449534	0.270658	0.365133	0.057206
차입금의존도	-0.565009	-0.144432	1.000000	-0.296486	-0.333407	0.023419	-0.242686
총부채회전율	0.418785	0.449534	-0.296486	1.000000	0.289369	0.592956	0.088937
총자본투자효율	0.407015	0.270658	-0.333407	0.289369	1.000000	0.124664	-0.010073
총자본회전율	-0.075471	0.365133	0.023419	0.592956	0.124664	1.000000	-0.042660
현금성자산비율	0.148976	0.057206	-0.242686	0.088937	-0.010073	-0.042660	1.000000



5-folds & Modeling

Reduced input features (13개)로 5-folds cross validation

- Random Forest
 - Adaboost
 - MLP (Deep Learning)
 - Ensemble (Voting)
- 기업별로 묶어서 random하게 기업이 7:3이 되도록 분할.
 - `X = df_matched[input_features[4:]]`
 - `X_reduced`
 - `y = df_matched[target_feature]`

```
In [26]: symbols = df_matched['Symbol'].values

# Symbol 기준 train/test 분할
unique_symbols = np.unique(symbols)
train_symbols, test_symbols = train_test_split(unique_symbols, test_size=0.3, ra

train_idx = df_matched['Symbol'].isin(train_symbols)
test_idx = df_matched['Symbol'].isin(test_symbols)

X_trainval = X_reduced[train_idx].reset_index(drop=True)
```

```

y_trainval = y[train_idx].reset_index(drop=True)
X_test = X_reduced[test_idx].reset_index(drop=True)
y_test = y[test_idx].reset_index(drop=True)

# 모델 정의
models = {
    'RandomForest': RandomForestClassifier(random_state=42),
    'AdaBoost': AdaBoostClassifier(random_state=42),
    'MLP': MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, random_state
}

```

Cross Validation

```

In [27]: best_models = {}

for model_name, model_template in models.items():
    print(f"\n🔍 모델: {model_name}")
    fold_scores = []

    best_model = None
    best_auc = -1
    best_fold = -1

    skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
    for fold, (train_idx_fold, val_idx_fold) in enumerate(skf.split(X_trainval,
        X_train = X_trainval.iloc[train_idx_fold]
        y_train = y_trainval.iloc[train_idx_fold]
        X_val = X_trainval.iloc[val_idx_fold]
        y_val = y_trainval.iloc[val_idx_fold]

        model = model_template.__class__(**model_template.get_params())
        model.fit(X_train, y_train)

        y_pred = model.predict(X_val)
        y_proba = model.predict_proba(X_val)[:, 1]

        acc = accuracy_score(y_val, y_pred)
        auc = roc_auc_score(y_val, y_proba)
        fold_scores.append((fold + 1, acc, auc))

    print(f"Fold {fold + 1}: Accuracy = {acc:.4f}, AUC = {auc:.4f}")

    if acc > 0.8 and auc > best_auc:
        best_model = model
        best_auc = auc
        best_fold = fold + 1

    if best_model is not None:
        best_models[model_name] = best_model
        print(f"선택된 Best Fold: Fold {best_fold} (AUC = {best_auc:.4f})\n")
    else:
        print(f"❌ 조건을 만족하는 fold 없음 (accuracy > 0.8)")

```

🔍 모델 : RandomForest

Fold 1: Accuracy = 0.8966, AUC = 0.9845
Fold 2: Accuracy = 0.9310, AUC = 0.9906
Fold 3: Accuracy = 0.9310, AUC = 0.9854
Fold 4: Accuracy = 0.8966, AUC = 0.9094
Fold 5: Accuracy = 0.9649, AUC = 0.9851
선택된 Best Fold: Fold 2 (AUC = 0.9906)

🔍 모델 : AdaBoost

Fold 1: Accuracy = 0.9310, AUC = 0.9420
Fold 2: Accuracy = 0.9310, AUC = 0.9896
Fold 3: Accuracy = 0.9310, AUC = 0.9813
Fold 4: Accuracy = 0.8966, AUC = 0.7354
Fold 5: Accuracy = 0.8947, AUC = 0.9606
선택된 Best Fold: Fold 2 (AUC = 0.9896)

🔍 모델 : MLP

Fold 1: Accuracy = 0.8621, AUC = 0.9207
Fold 2: Accuracy = 0.8448, AUC = 0.8812
Fold 3: Accuracy = 0.8966, AUC = 0.8917
Fold 4: Accuracy = 0.9310, AUC = 0.8771
Fold 5: Accuracy = 0.9123, AUC = 0.8319
선택된 Best Fold: Fold 1 (AUC = 0.9207)

```
c:\Users\HOME\Desktop\VSC\2025_spring_Data_Mining\DM_env\Lib\site-packages\sklearn\n\n\nneural_network\n\n_multilayer_perceptron.py:691: ConvergenceWarning: Stochastic Optimizer: Maximum iterations (500) reached and the optimization hasn't converged yet.\n\nwarnings.warn(
```

Test

```
In [28]: test_results = {}
voting_proba_list = []

for model_name, model in best_models.items():
    y_test_pred = model.predict(X_test)
    y_test_proba = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_test_pred)
    auc = roc_auc_score(y_test, y_test_proba)

    voting_proba_list.append(y_test_proba)

    test_results[model_name] = {
        'accuracy': acc,
        'roc_auc': auc
    }

# Soft Voting Ensemble (3개 모델 전부 있는 경우만)
if len(voting_proba_list) == 3:
    avg_proba = np.mean(voting_proba_list, axis=0)
    y_voting = (avg_proba >= 0.5).astype(int)

    acc_voting = accuracy_score(y_test, y_voting)
    auc_voting = roc_auc_score(y_test, avg_proba)
```

```

test_results['VotingEnsemble'] = {
    'accuracy': acc_voting,
    'roc_auc': auc_voting
}

# 결과 출력
for model_name, result in test_results.items():
    print(f"{model_name:15s} | Accuracy: {result['accuracy']:.4f} | ROC AUC: {re

```

RandomForest	Accuracy: 0.9160	ROC AUC: 0.9337
AdaBoost	Accuracy: 0.9237	ROC AUC: 0.8703
MLP	Accuracy: 0.9008	ROC AUC: 0.8473
VotingEnsemble	Accuracy: 0.9008	ROC AUC: 0.9460

Random Forest 기반 Feature Importance

```

In [29]: # 1. RandomForest 학습 (전체 데이터 사용)
rf = RandomForestClassifier(random_state=42)
rf.fit(X_reduced, y)

# 2. 중요도 추출
importances = rf.feature_importances_
feature_names = X_reduced.columns

# 3. 중요도 정리
importance_df = pd.DataFrame({
    'Feature': feature_names,
    'Importance': importances
}).sort_values(by='Importance', ascending=False)

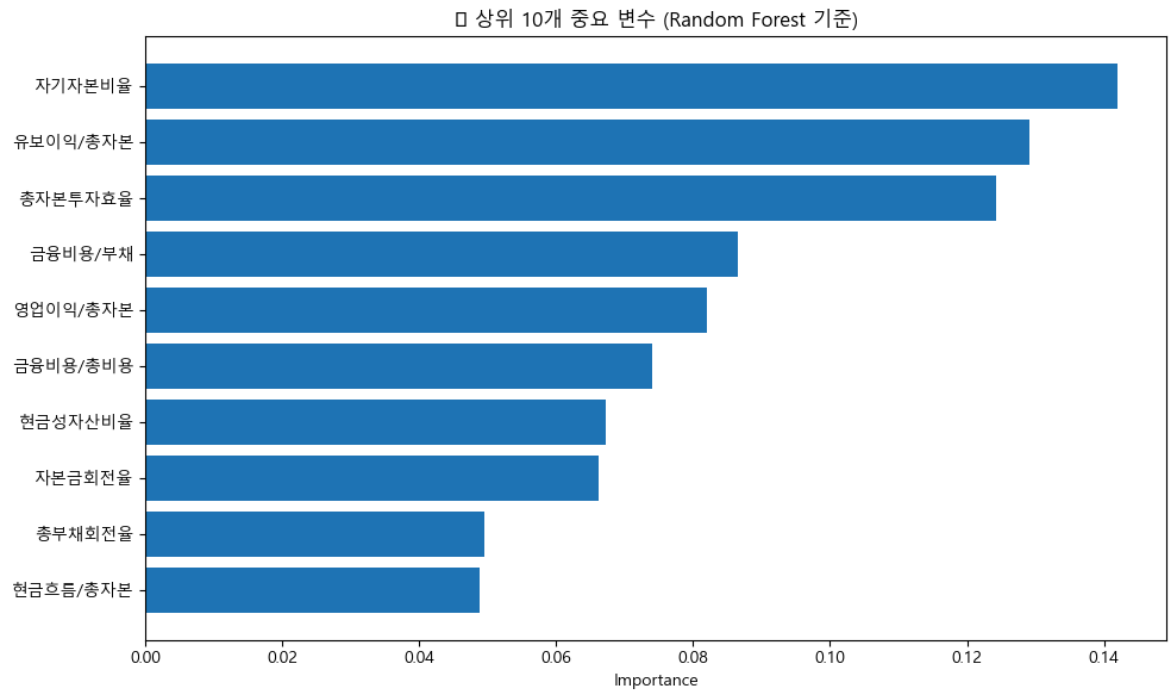
# 4. 상위 중요 변수 시각화
plt.figure(figsize=(10, 6))
plt.barh(importance_df['Feature'][:10][::-1], importance_df['Importance'][:10][:])
plt.title("🔥 상위 10개 중요 변수 (Random Forest 기준)")
plt.xlabel("Importance")
plt.tight_layout()
plt.show()

```

```

C:\Users\Public\Documents\ESTsoft\CreatorTemp\ipykernel_20236\1170696816.py:20: UserWarning: Glyph 128204 (\N{PUSHPIN}) missing from font(s) Malgun Gothic.
plt.tight_layout()
c:\Users\HOME\Desktop\VSC\2025_spring_Data_Mining\DM_env\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 128204 (\N{PUSHPIN}) missing from font(s) Malgun Gothic.
fig.canvas.print_figure(bytes_io, **kw)

```



Reduced Features 대신, 전체 Features로 modeling

- '총순자산수익률' 포함 14개

```
In [33]: symbols = df_matched['Symbol'].values

# Symbol 기준 train/test 분할
unique_symbols = np.unique(symbols)
train_symbols, test_symbols = train_test_split(unique_symbols, test_size=0.3, ra

train_idx = df_matched['Symbol'].isin(train_symbols)
test_idx = df_matched['Symbol'].isin(test_symbols)

X_trainval = X[train_idx].reset_index(drop=True)
y_trainval = y[train_idx].reset_index(drop=True)
X_test = X[test_idx].reset_index(drop=True)
y_test = y[test_idx].reset_index(drop=True)

# 모델 정의
models = {
    'RandomForest': RandomForestClassifier(random_state=42),
    'AdaBoost': AdaBoostClassifier(random_state=42),
    'MLP': MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, random_state
}
```

Cross Validation

```
In [34]: best_models = {}

for model_name, model_template in models.items():
    print(f"\n🔍 모델: {model_name}")
    fold_scores = []

    best_model = None
    best_auc = -1
    best_fold = -1
```

```

skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
for fold, (train_idx_fold, val_idx_fold) in enumerate(skf.split(X_trainval,
    X_train = X_trainval.iloc[train_idx_fold]
    y_train = y_trainval.iloc[train_idx_fold]
    X_val = X_trainval.iloc[val_idx_fold]
    y_val = y_trainval.iloc[val_idx_fold]

    model = model_template.__class__(**model_template.get_params())
    model.fit(X_train, y_train)

    y_pred = model.predict(X_val)
    y_proba = model.predict_proba(X_val)[:, 1]

    acc = accuracy_score(y_val, y_pred)
    auc = roc_auc_score(y_val, y_proba)
    fold_scores.append((fold + 1, acc, auc))

    print(f"Fold {fold + 1}: Accuracy = {acc:.4f}, AUC = {auc:.4f}")

    if acc > 0.8 and auc > best_auc:
        best_model = model
        best_auc = auc
        best_fold = fold + 1

    if best_model is not None:
        best_models[model_name] = best_model
        print(f"선택된 Best Fold: Fold {best_fold} (AUC = {best_auc:.4f})\n")
    else:
        print(f"❌ 조건을 만족하는 fold 없음 (accuracy > 0.8)")

```

🔍 모델: RandomForest

Fold 1: Accuracy = 0.9138, AUC = 0.9884
 Fold 2: Accuracy = 0.9310, AUC = 0.9740
 Fold 3: Accuracy = 0.9483, AUC = 0.9833
 Fold 4: Accuracy = 0.8966, AUC = 0.9531
 Fold 5: Accuracy = 0.9649, AUC = 0.9851
 선택된 Best Fold: Fold 1 (AUC = 0.9884)

🔍 모델: AdaBoost

Fold 1: Accuracy = 0.9310, AUC = 0.9749
 Fold 2: Accuracy = 0.9483, AUC = 0.9979
 Fold 3: Accuracy = 0.9138, AUC = 0.9354
 Fold 4: Accuracy = 0.8966, AUC = 0.7635
 Fold 5: Accuracy = 0.9123, AUC = 0.9511
 선택된 Best Fold: Fold 2 (AUC = 0.9979)

🔍 모델: MLP

Fold 1: Accuracy = 0.8621, AUC = 0.9130
 Fold 2: Accuracy = 0.9138, AUC = 0.9083
 Fold 3: Accuracy = 0.9138, AUC = 0.9396
 Fold 4: Accuracy = 0.9310, AUC = 0.8104
 Fold 5: Accuracy = 0.9123, AUC = 0.9340
 선택된 Best Fold: Fold 3 (AUC = 0.9396)

Test

```
In [35]: test_results = {}
voting_proba_list = []

for model_name, model in best_models.items():
    y_test_pred = model.predict(X_test)
    y_test_proba = model.predict_proba(X_test)[: , 1]

    acc = accuracy_score(y_test, y_test_pred)
    auc = roc_auc_score(y_test, y_test_proba)

    voting_proba_list.append(y_test_proba)

    test_results[model_name] = {
        'accuracy': acc,
        'roc_auc': auc
    }

# Soft Voting Ensemble (3개 모델 전부 있는 경우만)
if len(voting_proba_list) == 3:
    avg_proba = np.mean(voting_proba_list, axis=0)
    y_voting = (avg_proba >= 0.5).astype(int)

    acc_voting = accuracy_score(y_test, y_voting)
    auc_voting = roc_auc_score(y_test, avg_proba)

    test_results['VotingEnsemble'] = {
        'accuracy': acc_voting,
        'roc_auc': auc_voting
    }

# 결과 출력
for model_name, result in test_results.items():
    print(f"{model_name:15s} | Accuracy: {result['accuracy']:.4f} | ROC AUC: {re
```

RandomForest	Accuracy: 0.9008	ROC AUC: 0.9098
AdaBoost	Accuracy: 0.8855	ROC AUC: 0.8332
MLP	Accuracy: 0.8931	ROC AUC: 0.8665
VotingEnsemble	Accuracy: 0.8931	ROC AUC: 0.9225

Random Forest 기반 Feature Importance

```
In [37]: # 1. RandomForest 학습 (전체 데이터 사용)
rf = RandomForestClassifier(random_state=42)
rf.fit(X, y)

# 2. 중요도 추출
importances = rf.feature_importances_
feature_names = X.columns

# 3. 중요도 정리
importance_df = pd.DataFrame({
    'Feature': feature_names,
    'Importance': importances
}).sort_values(by='Importance', ascending=False)

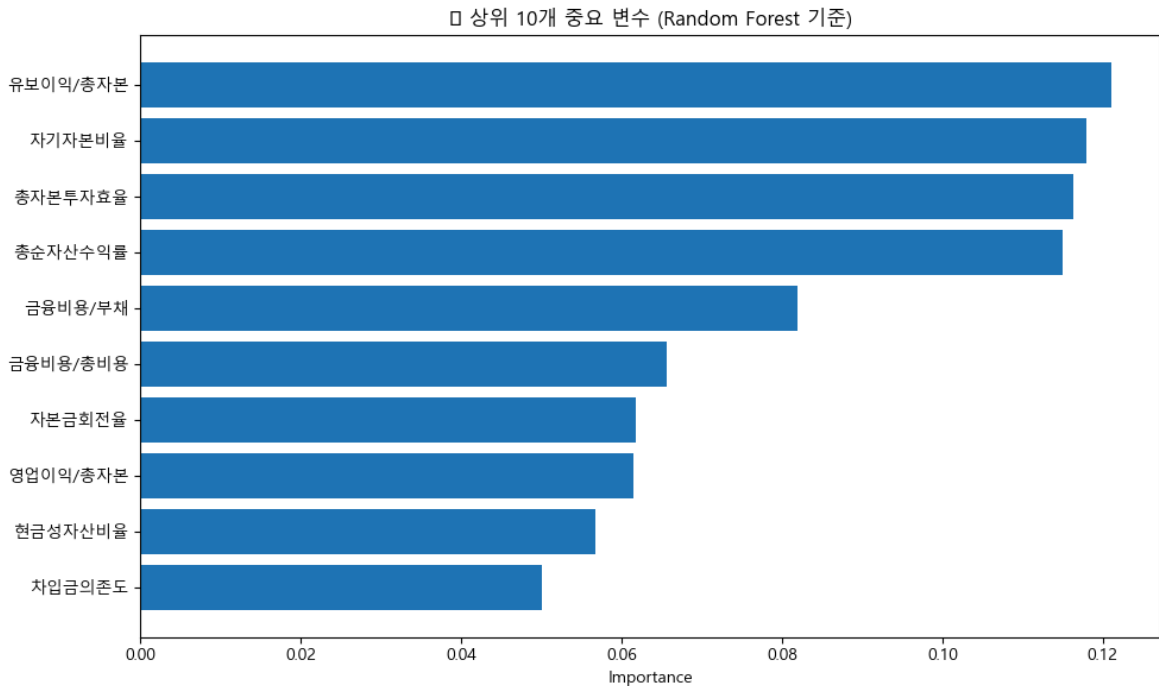
# 4. 상위 중요 변수 시각화
plt.figure(figsize=(10, 6))
plt.barh(importance_df['Feature'][:10][::-1], importance_df['Importance'][:10][::-1])
plt.title("💖 상위 10개 중요 변수 (Random Forest 기준)")
```

```
plt.xlabel("Importance")
plt.tight_layout()
plt.show()
```

C:\Users\Public\Documents\ESTsoft\CreatorTemp\ipykernel_20236\3014796290.py:20: UserWarning: Glyph 128204 (\N{PUSHPIN}) missing from font(s) Malgun Gothic.

plt.tight_layout()
c:\Users\HOME\Desktop\VSC\2025_spring_Data_Mining\DM_env\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 128204 (\N{PUSHPIN}) missing from font(s) Malgun Gothic.

```
fig.canvas.print_figure(bytes_io, **kw)
```



- Reduced Features의 Feature Importance에 비해 Feature Importance가 일부 순서가 변동되었으나, 크게 달라지지는 않았습니다.

성능 비교

- Reduced Features vs 전체 Features

```
In [39]: # 모델 정의
models = {
    'RandomForest': RandomForestClassifier(random_state=42),
    'AdaBoost': AdaBoostClassifier(random_state=42),
    'MLP': MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, random_state=42)
}

# 평가할 feature set
X_variants = {
    'AllFeatures': X,
    'ReducedFeatures': X_reduced
}

# Symbol 기반 train/test 분할
symbols = df_matched['Symbol'].values
unique_symbols = np.unique(symbols)
train_symbols, test_symbols = train_test_split(unique_symbols, test_size=0.3, random_state=42)
```

```

# 결과 저장
results = {}

# 각 feature set 별로 test 평가
for variant_name, X_variant in X_variants.items():
    print(f"\n===== 🌟 {variant_name} 🌟 =====")

    X_trainval = X_variant[df_matched['Symbol'].isin(train_symbols)].reset_index
    y_trainval = y[df_matched['Symbol'].isin(train_symbols)].reset_index(drop=True)
    X_test = X_variant[df_matched['Symbol'].isin(test_symbols)].reset_index(drop=True)
    y_test = y[df_matched['Symbol'].isin(test_symbols)].reset_index(drop=True)

    best_models = {}
    for model_name, model_template in models.items():
        best_model = None
        best_auc = -1
        best_fold = -1

        skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
        for fold, (train_idx, val_idx) in enumerate(skf.split(X_trainval, y_trainval)):
            X_train = X_trainval.iloc[train_idx]
            y_train = y_trainval.iloc[train_idx]
            X_val = X_trainval.iloc[val_idx]
            y_val = y_trainval.iloc[val_idx]

            model = model_template.__class__(**model_template.get_params())
            model.fit(X_train, y_train)

            y_pred = model.predict(X_val)
            y_proba = model.predict_proba(X_val)[:, 1]
            acc = accuracy_score(y_val, y_pred)
            auc = roc_auc_score(y_val, y_proba)

            if acc > 0.8 and auc > best_auc:
                best_model = model
                best_auc = auc
                best_fold = fold + 1

        if best_model is not None:
            best_models[model_name] = best_model
        else:
            print(f"❌ {variant_name} - {model_name}: No valid fold")

# test 평가
test_result = {}
proba_list = []

for model_name, model in best_models.items():
    y_test_pred = model.predict(X_test)
    y_test_proba = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_test_pred)
    auc = roc_auc_score(y_test, y_test_proba)

    test_result[model_name] = {
        'Accuracy': round(acc, 4),
        'ROC_AUC': round(auc, 4)
    }

    proba_list.append(y_test_proba)

```

```

# Soft Voting Ensemble
if len(proba_list) == 3:
    avg_proba = np.mean(proba_list, axis=0)
    y_vote = (avg_proba >= 0.5).astype(int)
    acc_vote = accuracy_score(y_test, y_vote)
    auc_vote = roc_auc_score(y_test, avg_proba)

    test_result['VotingEnsemble'] = {
        'Accuracy': round(acc_vote, 4),
        'ROC_AUC': round(auc_vote, 4)
    }

    results[variant_name] = pd.DataFrame(test_result).T

# 결과 비교 출력
print("\n🔗 Test 성능 비교 요약:")
comparison = pd.concat(results, axis=1)
print(comparison)

```

===== 🌟 AllFeatures 🌟 =====

===== 🌟 ReducedFeatures 🌟 =====

🔗 Test 성능 비교 요약:

	AllFeatures		ReducedFeatures	
	Accuracy	ROC_AUC	Accuracy	ROC_AUC
RandomForest	0.9008	0.9098	0.9160	0.9337
AdaBoost	0.8855	0.8332	0.9237	0.8703
MLP	0.8931	0.8665	0.9008	0.8473
VotingEnsemble	0.8931	0.9225	0.9008	0.9460

c:\Users\HOME\Desktop\VSC\2025_spring_Data_Mining\DM_env\Lib\site-packages\sklearn\n\n\nneural_network\multilayer_perceptron.py:691: ConvergenceWarning: Stochastic Optimizer: Maximum iterations (500) reached and the optimization hasn't converged yet.

warnings.warn(

- 성능이 모두 훌륭하네요. 전반적으로는 '총순자산수익률'의 포함 여부가 성능에 영향을 미치지 않는 것 같습니다. 빼는 게 더 simple model이 될 것 같아요.
- ** pre_modeling_ver1.ipynb ** 에 비해 성능이 매우 개선되었습니다. (Below: 변화된 점)
 - 1. 상장폐지 이후 data 삭제
 - 2. 전처리 정제
 - 3. matched-pairs sampling 기법 도입. (이전: Simple case sampling)

3. Time Series Analysis

⇒ label의 경우 sliding window 방식으로 처리.

- window size = k
- N-k, ..., N-1 recent k개 년도 data → N 년도 bad = 1 여부.
- each sample = (N-k, ..., N-1 recent k개 년도 input feature & N 년도 label)
- dataset = all samples, 기업별 구분 무시 (k개 년도에 따른 label값만 중요)

- k개 년도 연속 sliding window 생성
 - 불연속 window 제거: 연도가 연속적이지 않은 경우에는 해당 sample 제거. (ex. 2000, **2001, 2003**, 2004, 2005)
 - each sample마다 y-label 부여.
- "<https://gmnam.tistory.com/230>"에 나오는 TimeSeriesSplit 방식 사용.
 - Time Series data의 5folds 방식, 시간의 연속성 때문
- k개 년도 sliding window 생성

```
In [31]: print(f"df's columns: {columns}")
print(f"input features: {input_features}")
print(f"target feature: {target_feature}")
```

```
df's columns: ['Symbol', '회계년', '상장폐지년도', 'duration', '금융비용/부채',
'금융비용/총비용', '설비투자효율', '현금흐름/총자본', '영업이익/총자본', '유보이익/총자본', '자기자본비율', '자본금회전율', '차입금의존도', '총부채회전율', '총자본투자효율', '총자본회전율', '총순자산수익률', '현금성자산비율', 'Bad']
input features: ['Symbol', '회계년', '상장폐지년도', 'duration', '금융비용/부채',
'금융비용/총비용', '설비투자효율', '현금흐름/총자본', '영업이익/총자본', '유보이익/총자본', '자기자본비율', '자본금회전율', '차입금의존도', '총부채회전율', '총자본투자효율', '총자본회전율', '총순자산수익률', '현금성자산비율']
target feature: Bad
```

```
In [40]: selected_features = [ input_features[1] ] + input_features[4:-1]

def generate_sliding_windows(df, k):
    samples = []
    grouped = df.groupby('Symbol')

    for symbol, group in grouped:
        group = group.sort_values('회계년').reset_index(drop=True)
        years = group['회계년'].values
        features = group[selected_features].values

        for i in range(len(group) - k):
            window_years = years[i:i + k + 1]
            if not np.all(np.diff(window_years) == 1): # 연속 연도 아닌 경우 제
                continue

            X_window = features[i:i + k] # k개 년도의 연속된 feature
            prediction_year = years[i + k] # 예측할 연도
            y_label = group.loc[i + k, 'Bad'] # 예측할 연도의 label

            samples.append((X_window, y_label, prediction_year))

    return samples
```

```
In [41]: k = 3 # tuning parameter, k=5는 data가 너무 적어서
samples = generate_sliding_windows(df, k) # X_window, y_label, prediction_year

# prediction_year 기준으로 정렬
samples.sort(key=lambda x: x[2])
```

Label Distribution (Case Sampling)

```
In [43]: def check_label_distribution(dataset, name="Dataset"):
# (X, y) 또는 (X, y, year) 모두 처리 가능
labels = [y for item in dataset for i, y in enumerate(item) if i == 1]
unique, counts = np.unique(labels, return_counts=True)

print(f"\n{name} 샘플 수: {len(labels)}")
for u, c in zip(unique, counts):
    print(f"- Label {u}: {c}개 ({c / len(labels):.2%}")

check_label_distribution(samples)
```

Dataset 샘플 수: 79390
- Label 0: 79332개 (99.93%)
- Label 1: 58개 (0.07%)

- 위에서 time sequence로 분석하기 전 Label 1: 70개 였는데, 그 중 12개가 사라졌습니다. 아무래도 연속되지 않은 연도가 12개가 있었던 것 같아요
- 이전에 제가 한 NA처리 방식과 다르게 NA가 처리되니, 더 적은 sample이 살아남았습니다.
- 이전에는 (fin_modeling_ver2.ipynb) 8069 중 6718, 1351개 살아남았어요
- sample: X_window, y_label, prediction_year

```
In [91]: print(f"1st sample example: \n{samples[0]}\n")

print(f"1st sample input: {samples[0][0]}")
print(f"1st sample label: {samples[0][1]}")
if samples[0][1] == 1:
    print(f"1st sample 상장폐지년도: {samples[0][2]}")
else:
    print(f"1st sample: 상장폐지 X")
print(f"1st sample 입력연도: {samples[0][2]-k}-{samples[0][2]-(k-2)}")
print(f"1st sample 예측연도: {samples[0][2]}")

print([s[2] for s in samples[:10]]) # 처음 10개 예측 연도 출력
```

1st sample example:

```
(array([[ 1.99900000e+03,  0.00000000e+00,  0.00000000e+00,
         2.49356708e+00,  1.12994237e-01, -4.17364802e-01,
        -4.69007612e-01,  4.67222215e-02,  1.06279454e+00,
         0.00000000e+00,  8.06462340e-02, -1.95002108e-02,
         7.68782628e-02, -3.37670871e-01],
        [ 2.00000000e+03,  0.00000000e+00,  0.00000000e+00,
         2.85387228e+00,  7.20504946e-01,  7.89422480e-02,
        -3.87288803e-01,  4.11040056e-02,  1.15143469e+00,
         0.00000000e+00,  7.97274734e-02,  3.24484260e-03,
         7.64503549e-02,  4.62677301e-02],
        [ 2.00100000e+03,  0.00000000e+00,  0.00000000e+00,
         2.66806945e+00,  6.43380103e-01,  3.10514607e-01,
        -1.33537372e-01,  4.52846104e-02,  1.06001783e+00,
         0.00000000e+00,  6.65939849e-02,  1.40615330e-02,
         6.35783022e-02,  2.05989545e-01]]), np.int64(0), np.int64(2002))
```

1st sample input: [[1.99900000e+03 0.00000000e+00 0.00000000e+00 2.49356708e+00

```
1.12994237e-01 -4.17364802e-01 -4.69007612e-01  4.67222215e-02
1.06279454e+00  0.00000000e+00  8.06462340e-02 -1.95002108e-02
7.68782628e-02 -3.37670871e-01]
[ 2.00000000e+03  0.00000000e+00  0.00000000e+00  2.85387228e+00
 7.20504946e-01  7.89422480e-02 -3.87288803e-01  4.11040056e-02
 1.15143469e+00  0.00000000e+00  7.97274734e-02  3.24484260e-03
 7.64503549e-02  4.62677301e-02]
[ 2.00100000e+03  0.00000000e+00  0.00000000e+00  2.66806945e+00
 6.43380103e-01  3.10514607e-01 -1.33537372e-01  4.52846104e-02
 1.06001783e+00  0.00000000e+00  6.65939849e-02  1.40615330e-02
 6.35783022e-02  2.05989545e-01]]
```

1st sample label: 0

1st sample: 상장폐지 X

1st sample 입력연도: 1999-2001

1st sample 예측연도: 2002

```
[np.int64(2002), np.int64(2002), np.int64(2002), np.int64(2002), np.int64(2002),
 np.int64(2002), np.int64(2002), np.int64(2002), np.int64(2002), np.int64(2002)]
```

- 배열로 변환: TimeSeriesSplit (Time-Sequence)
- Test set 분리: 2017~2024년
- Modeling Set: 1999+ k ~ 2016년

```
In [76]: X_window = np.array([s[0] for s in samples])
y_label = np.array([s[1] for s in samples])
prediction_year = np.array([s[2] for s in samples])

test_mask = (prediction_year >= 2017)
prediction_year_modeling = prediction_year[~test_mask]

X_test = X_window[test_mask]
y_test = y_label[test_mask]

X_modeling = X_window[~test_mask]
y_modeling = y_label[~test_mask]
```

Case sampling: y label ratio 1:1

- TimeSeriesSplit 도중 loss function에 weight를 부여해서 불균형을 해소할 예정입니다.

```
In [52]: def check_label_distribution_from_array(y, name=""):
y = np.array(y)
total = len(y)
num_1 = np.sum(y == 1)
num_0 = np.sum(y == 0)
print(f"[{name}] Total: {total}, 1 비율: {num_1/total:.4f}, 1 개수: {num_1},

check_label_distribution_from_array(y_modeling, "Modeling Set")
check_label_distribution_from_array(y_test, "Test Set")
```

[Modeling Set] Total: 50875, 1 비율: 0.0009, 1 개수: 48, 0 개수: 50827
[Test Set] Total: 28515, 1 비율: 0.0004, 1 개수: 10, 0 개수: 28505

Modeling

- Accuracy와 함께 AUC curve를 사용. (Un-balanced ratio data)
- Test Set: Accuracy > 0.8 & AUC가 가장 큰 fold의 model 선택
- Seed: 고정, 결과 재현성 확보

```
In [56]: def set_seed(seed=42):
random.seed(seed)
np.random.seed(seed)
torch.manual_seed(seed)
torch.cuda.manual_seed_all(seed)
# CuDNN 관련 추가 옵션 (완전한 재현성)
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False
```

```
In [59]: set_seed(42)
```

- MLP

```
In [60]: class MLP(nn.Module):
def __init__(self, input_size):
super().__init__()
self.fc = nn.Sequential(
nn.Flatten(),
nn.Linear(input_size, 128),
nn.BatchNorm1d(128),
nn.ReLU(),
nn.Dropout(0.3),
nn.Linear(128, 64),
nn.BatchNorm1d(64),
nn.ReLU(),
nn.Dropout(0.3),
nn.Linear(64, 1)
)

def forward(self, x):
```

```

        return self.fc(x)

# hyperparameters
n_splits = 5
batch_size = 64
lr = 1e-3
num_epochs = 20

input_size = X_modeling.shape[1] * X_modeling.shape[2]

```

Cross Validation

```

In [77]: MLP_best_fold_info = {
    "fold": None,
    "acc": 0,
    "auc": 0,
    "model": None
}

MLP_acc_list = []
MLP_auc_list = []

tscv = TimeSeriesSplit(n_splits=n_splits)

print("==== TimeSeriesSplit MLP Training with Weighted Loss ====")
for fold, (train_idx, val_idx) in enumerate(tscv.split(X_modeling)):
    # 데이터 선택
    X_train = torch.tensor(X_modeling[train_idx], dtype=torch.float32)
    y_train = torch.tensor(y_modeling[train_idx], dtype=torch.float32)
    X_val = torch.tensor(X_modeling[val_idx], dtype=torch.float32)
    y_val = torch.tensor(y_modeling[val_idx], dtype=torch.float32)

    # 연도 출력
    train_years = prediction_year_modeling[train_idx]
    val_years = prediction_year_modeling[val_idx]
    print(f"[Fold {fold+1}] 예측 연도 범위 (Train): {train_years.min()} ~ {train_years.max()}")

    # 불균형 보정
    num_pos = (y_train == 1).sum().item()
    num_neg = (y_train == 0).sum().item()
    pos_weight = num_neg / num_pos if num_pos > 0 else 1.0
    pos_weight_tensor = torch.tensor([pos_weight], dtype=torch.float32)

    # 모델 정의
    model = MLP(input_size)
    criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight_tensor)
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)
    train_loader = DataLoader(TensorDataset(X_train, y_train), batch_size=batch_size)

    # 학습
    for epoch in range(num_epochs):
        model.train()
        for xb, yb in train_loader:
            optimizer.zero_grad()
            logits = model(xb).squeeze()
            loss = criterion(logits, yb)
            loss.backward()
            optimizer.step()

```

```

# 검증
model.eval()
with torch.no_grad():
    val_logits = model(X_val).squeeze()
    val_probs = torch.sigmoid(val_logits).numpy()
    val_pred = (val_probs >= 0.5).astype(int)

    acc = accuracy_score(y_val.numpy(), val_pred)
    auc = roc_auc_score(y_val.numpy(), val_probs)

    MLP_acc_list.append(acc)
    MLP_auc_list.append(auc)

    print(f"[Fold {fold+1}] Validation Accuracy: {acc:.4f}, ROC AUC: {auc:.4f}")

    if acc > 0.8 and auc > MLP_best_fold_info["auc"]:
        MLP_best_fold_info.update({
            "fold": fold + 1,
            "acc": acc,
            "auc": auc,
            "model": model
        })

```

```

==== TimeSeriesSplit MLP Training with Weighted Loss ====
[Fold 1] 예측 연도 범위 (Train): 2002 ~ 2004, (Val): 2004 ~ 2007
[Fold 1] Validation Accuracy: 0.9993, ROC AUC: 0.3759

[Fold 2] 예측 연도 범위 (Train): 2002 ~ 2007, (Val): 2007 ~ 2009
[Fold 2] Validation Accuracy: 0.9912, ROC AUC: 0.9263

[Fold 3] 예측 연도 범위 (Train): 2002 ~ 2009, (Val): 2009 ~ 2012
[Fold 3] Validation Accuracy: 0.9700, ROC AUC: 0.7379

[Fold 4] 예측 연도 범위 (Train): 2002 ~ 2012, (Val): 2012 ~ 2014
[Fold 4] Validation Accuracy: 0.9995, ROC AUC: 0.5099

[Fold 5] 예측 연도 범위 (Train): 2002 ~ 2014, (Val): 2014 ~ 2016
[Fold 5] Validation Accuracy: 0.9968, ROC AUC: 0.5660

```

Test

```

In [ ]: X_test_tensor = torch.tensor(X_test, dtype=torch.float32)
        y_test_tensor = torch.tensor(y_test, dtype=torch.float32)

        print("==== Test Evaluation ====")
        if MLP_best_fold_info["model"] is not None:
            best_model = MLP_best_fold_info["model"]
            best_model.eval()
            with torch.no_grad():
                test_logits = best_model(X_test_tensor).squeeze()
                test_probs = torch.sigmoid(test_logits).numpy()
                test_pred = (test_probs >= 0.5).astype(int)

                test_acc = accuracy_score(y_test_tensor.numpy(), test_pred)
                test_auc = roc_auc_score(y_test_tensor.numpy(), test_probs)

                print(f"[Selected Fold {MLP_best_fold_info['fold']}] Test Accuracy: {test_acc:.4f}, ROC AUC: {test_auc:.4f}")
            else:
                print("No fold satisfied the accuracy > 0.8 condition.")

```

==== Test Evaluation ====

[Selected Fold 2] Test Accuracy: 0.9991, ROC AUC: 0.6820

- GRU

```
In [80]: class GRUModel(nn.Module):
    def __init__(self, input_size, hidden_size=64, num_layers=1, bidirectional=False):
        super().__init__()
        self.gru = nn.GRU(
            input_size=input_size,
            hidden_size=hidden_size,
            num_layers=num_layers,
            batch_first=True,
            bidirectional=bidirectional
        )
        self.fc = nn.Linear(hidden_size * (2 if bidirectional else 1), 1)

    def forward(self, x):
        # x: (batch, seq_len, input_size)
        out, _ = self.gru(x)
        # 마지막 타임스텝의 출력만 사용
        out = out[:, -1, :] # shape: (batch, hidden_size)
        return self.fc(out) # logits (no sigmoid)

# hyperparameters
n_splits = 5
batch_size = 64
lr = 1e-3
num_epochs = 20

input_dim = X_modeling.shape[2] # 시계열 feature 수
```

Cross Validation

```
In [81]: GRU_best_fold_info = {
    "fold": None,
    "acc": 0,
    "auc": 0,
    "model": None
}

GRU_acc_list = []
GRU_auc_list = []

tscv = TimeSeriesSplit(n_splits=n_splits)

print("==== TimeSeriesSplit GRU Training with Weighted Loss ====")
for fold, (train_idx, val_idx) in enumerate(tscv.split(X_modeling)):
    X_train = torch.tensor(X_modeling[train_idx], dtype=torch.float32)
    y_train = torch.tensor(y_modeling[train_idx], dtype=torch.float32)
    X_val = torch.tensor(X_modeling[val_idx], dtype=torch.float32)
    y_val = torch.tensor(y_modeling[val_idx], dtype=torch.float32)

    train_years = prediction_year_modeling[train_idx]
    val_years = prediction_year_modeling[val_idx]
    print(f"[Fold {fold+1}] 입력 연도 범위 (Train): {train_years.min()} ~ {train_years.max()}")
```

```

num_pos = (y_train == 1).sum().item()
num_neg = (y_train == 0).sum().item()
pos_weight = num_neg / num_pos if num_pos > 0 else 1.0
pos_weight_tensor = torch.tensor([pos_weight], dtype=torch.float32)

model = GRUModel(input_size=X_modeling.shape[2]) # (N, T, D)
criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight_tensor)
optimizer = torch.optim.Adam(model.parameters(), lr=lr)

train_loader = DataLoader(TensorDataset(X_train, y_train), batch_size=batch_

for epoch in range(1, num_epochs + 1):
    model.train()
    for xb, yb in train_loader:
        optimizer.zero_grad()
        output = model(xb).squeeze()
        loss = criterion(output, yb)
        loss.backward()
        optimizer.step()

model.eval()
with torch.no_grad():
    val_logits = model(X_val).squeeze()
    val_probs = torch.sigmoid(val_logits).numpy()
    val_pred = (val_probs >= 0.5).astype(int)

    acc = accuracy_score(y_val.numpy(), val_pred)
    auc = roc_auc_score(y_val.numpy(), val_probs)

    GRU_acc_list.append(acc)
    GRU_auc_list.append(auc)

    print(f"[Fold {fold+1}] Validation Accuracy: {acc:.4f}, ROC AUC: {auc:.4f}")

    if acc > 0.8 and auc > GRU_best_fold_info["auc"]:
        GRU_best_fold_info.update({
            "fold": fold + 1,
            "acc": acc,
            "auc": auc,
            "model": model
        })

```

==== TimeSeriesSplit GRU Training with Weighted Loss ====

[Fold 1] 입력 연도 범위 (Train): 2002 ~ 2004, (Val): 2004 ~ 2007
[Fold 1] Validation Accuracy: 0.9994, ROC AUC: 0.5048

[Fold 2] 입력 연도 범위 (Train): 2002 ~ 2007, (Val): 2007 ~ 2009
[Fold 2] Validation Accuracy: 0.9995, ROC AUC: 0.4979

[Fold 3] 입력 연도 범위 (Train): 2002 ~ 2009, (Val): 2009 ~ 2012
[Fold 3] Validation Accuracy: 0.9983, ROC AUC: 0.4996

[Fold 4] 입력 연도 범위 (Train): 2002 ~ 2012, (Val): 2012 ~ 2014
[Fold 4] Validation Accuracy: 0.9995, ROC AUC: 0.5082

[Fold 5] 입력 연도 범위 (Train): 2002 ~ 2014, (Val): 2014 ~ 2016
[Fold 5] Validation Accuracy: 0.9996, ROC AUC: 0.5008

Test

```
In [82]: X_test_tensor = torch.tensor(X_test, dtype=torch.float32)
y_test_tensor = torch.tensor(y_test, dtype=torch.float32)

print("==== GRU Test Evaluation ====")
if GRU_best_fold_info["model"] is not None:
    best_model = GRU_best_fold_info["model"]
    best_model.eval()
    with torch.no_grad():
        test_logits = best_model(X_test_tensor).squeeze()
        test_probs = torch.sigmoid(test_logits).numpy()
        test_pred = (test_probs >= 0.5).astype(int)

        test_acc = accuracy_score(y_test_tensor.numpy(), test_pred)
        test_auc = roc_auc_score(y_test_tensor.numpy(), test_probs)

        print(f"[Selected Fold {GRU_best_fold_info['fold']}] Test Accuracy: {tes
else:
    print("No fold satisfied the accuracy > 0.8 condition.")
```

```
==== GRU Test Evaluation ====
[Selected Fold 4] Test Accuracy: 0.9996, ROC AUC: 0.5131
```

- TCN

```
In [83]: class Chomp1d(nn.Module):
    """Conv1d의 padding 때문에 생기는 output 뒤쪽 값을 잘라내는 역할"""
    def __init__(self, chomp_size):
        super().__init__()
        self.chomp_size = chomp_size

    def forward(self, x):
        return x[:, :, :-self.chomp_size]

class TemporalBlock(nn.Module):
    def __init__(self, in_channels, out_channels, kernel_size, stride, dilation,
        super().__init__()
        self.net = nn.Sequential(
            nn.Conv1d(in_channels, out_channels, kernel_size, stride=stride,
                padding=padding, dilation=dilation),
            Chomp1d(padding),
            nn.BatchNorm1d(out_channels),
            nn.ReLU(),
            nn.Dropout(dropout),
            nn.Conv1d(out_channels, out_channels, kernel_size, stride=stride,
                padding=padding, dilation=dilation),
            Chomp1d(padding),
            nn.BatchNorm1d(out_channels),
            nn.ReLU(),
            nn.Dropout(dropout)
        )
        self.downsample = nn.Conv1d(in_channels, out_channels, 1) if in_channels
        self.relu = nn.ReLU()

    def forward(self, x):
        out = self.net(x)
        res = x if self.downsample is None else self.downsample(x)
        return self.relu(out + res)
```

```

class TCN(nn.Module):
    def __init__(self, input_size, num_channels, kernel_size=2, dropout=0.2):
        super().__init__()
        layers = []
        num_levels = len(num_channels)
        for i in range(num_levels):
            dilation_size = 2 ** i
            in_ch = input_size if i == 0 else num_channels[i - 1]
            out_ch = num_channels[i]
            layers += [TemporalBlock(in_ch, out_ch, kernel_size, stride=1,
                                    dilation=dilation_size,
                                    padding=(kernel_size - 1) * dilation_size,
                                    dropout=dropout)]

        self.tcn = nn.Sequential(*layers)
        self.fc = nn.Linear(num_channels[-1], 1)

    def forward(self, x):
        # x: (batch, seq_len, input_dim)
        x = x.permute(0, 2, 1) # → (batch, input_dim, seq_len)
        y = self.tcn(x)        # → (batch, out_ch, seq_len)
        y = y[:, :, -1]        # 마지막 타임스텝만
        return self.fc(y)      # logits 출력

```

Cross Validation

```

In [87]: TCN_best_fold_info = {
    "fold": None,
    "acc": 0,
    "auc": 0,
    "model": None
}

TCN_acc_list = []
TCN_auc_list = []

tscv = TimeSeriesSplit(n_splits=n_splits)

print("==== TimeSeriesSplit TCN Training with Weighted Loss ====")
for fold, (train_idx, val_idx) in enumerate(tscv.split(X_modeling)):
    X_train = torch.tensor(X_modeling[train_idx], dtype=torch.float32)
    y_train = torch.tensor(y_modeling[train_idx], dtype=torch.float32)
    X_val = torch.tensor(X_modeling[val_idx], dtype=torch.float32)
    y_val = torch.tensor(y_modeling[val_idx], dtype=torch.float32)

    train_years = prediction_year_modeling[train_idx]
    val_years = prediction_year_modeling[val_idx]
    print(f"[Fold {fold+1}] 입력 연도 범위 (Train): {train_years.min()} ~ {train_years.max()}")

    num_pos = (y_train == 1).sum().item()
    num_neg = (y_train == 0).sum().item()
    pos_weight = num_neg / num_pos if num_pos > 0 else 1.0
    pos_weight_tensor = torch.tensor([pos_weight], dtype=torch.float32)

    model = TCN(input_size=X_modeling.shape[2], num_channels=[64, 64, 64], kernel_size=2,
                 criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight_tensor),
                 optimizer = torch.optim.Adam(model.parameters()), lr=lr)

    train_loader = DataLoader(TensorDataset(X_train, y_train), batch_size=batch_size)

```

```

for epoch in range(1, num_epochs + 1):
    model.train()
    for xb, yb in train_loader:
        optimizer.zero_grad()
        output = model(xb).squeeze()
        loss = criterion(output, yb)
        loss.backward()
        optimizer.step()

    model.eval()
    with torch.no_grad():
        val_logits = model(X_val).squeeze()
        val_probs = torch.sigmoid(val_logits).numpy()
        val_pred = (val_probs >= 0.5).astype(int)

        acc = accuracy_score(y_val.numpy(), val_pred)
        auc = roc_auc_score(y_val.numpy(), val_probs)

        TCN_acc_list.append(acc)
        TCN_auc_list.append(auc)

    print(f"[Fold {fold+1}] Validation Accuracy: {acc:.4f}, ROC AUC: {auc:.4f}")

    if acc > 0.8 and auc > TCN_best_fold_info["auc"]:
        TCN_best_fold_info.update({
            "fold": fold + 1,
            "acc": acc,
            "auc": auc,
            "model": model
        })

```

==== TimeSeriesSplit TCN Training with Weighted Loss ====

[Fold 1] 입력 연도 범위 (Train): 2002 ~ 2004, (Val): 2004 ~ 2007

[Fold 1] Validation Accuracy: 0.9919, ROC AUC: 0.4705

[Fold 2] 입력 연도 범위 (Train): 2002 ~ 2007, (Val): 2007 ~ 2009

[Fold 2] Validation Accuracy: 0.9993, ROC AUC: 0.8990

[Fold 3] 입력 연도 범위 (Train): 2002 ~ 2009, (Val): 2009 ~ 2012

[Fold 3] Validation Accuracy: 0.9940, ROC AUC: 0.7806

[Fold 4] 입력 연도 범위 (Train): 2002 ~ 2012, (Val): 2012 ~ 2014

[Fold 4] Validation Accuracy: 0.9992, ROC AUC: 0.5919

[Fold 5] 입력 연도 범위 (Train): 2002 ~ 2014, (Val): 2014 ~ 2016

[Fold 5] Validation Accuracy: 0.9992, ROC AUC: 0.4825

Test

```

In [88]: X_test_tensor = torch.tensor(X_test, dtype=torch.float32)
         y_test_tensor = torch.tensor(y_test, dtype=torch.float32)

         print("==== TCN Test Evaluation ====")
         if TCN_best_fold_info["model"] is not None:
             best_model = TCN_best_fold_info["model"]
             best_model.eval()
             with torch.no_grad():
                 test_logits = best_model(X_test_tensor).squeeze()

```

```

test_probs = torch.sigmoid(test_logits).numpy()
test_pred = (test_probs >= 0.5).astype(int)

test_acc = accuracy_score(y_test_tensor.numpy(), test_pred)
test_auc = roc_auc_score(y_test_tensor.numpy(), test_probs)

print(f"[Selected Fold {TCN_best_fold_info['fold']}] Test Accuracy: {tes
else:
print("No fold satisfied the accuracy > 0.8 condition.")

```

==== TCN Test Evaluation ====

[Selected Fold 2] Test Accuracy: 0.9996, ROC AUC: 0.7458

- Ensemble in Test
 - Soft Voting
 - Weighted Voting

```

In [89]: # Test data tensor
X_test_tensor = torch.tensor(X_test, dtype=torch.float32)
y_test_tensor = torch.tensor(y_test, dtype=torch.float32)

# 개별 모델 확률 예측
with torch.no_grad():
    mlp_probs = torch.sigmoid(MLP_best_fold_info["model"](X_test_tensor)).squeeze
    gru_probs = torch.sigmoid(GRU_best_fold_info["model"](X_test_tensor)).squeeze
    tcn_probs = torch.sigmoid(TCN_best_fold_info["model"](X_test_tensor)).squeeze

# ===== 1. Soft Voting =====
ensemble_probs_soft = (mlp_probs + gru_probs + tcn_probs) / 3
ensemble_pred_soft = (ensemble_probs_soft >= 0.5).astype(int)

acc_soft = accuracy_score(y_test_tensor.numpy(), ensemble_pred_soft)
auc_soft = roc_auc_score(y_test_tensor.numpy(), ensemble_probs_soft)

print(f"(1) [Soft Voting Ensemble] Test Accuracy: {acc_soft:.4f}, ROC AUC: {auc_

# ===== 2. Weighted Voting (AUC 기반 가중치) =====
auc_sum = MLP_best_fold_info["auc"] + GRU_best_fold_info["auc"] + TCN_best_fold_
w_mlp = MLP_best_fold_info["auc"] / auc_sum
w_gru = GRU_best_fold_info["auc"] / auc_sum
w_tcn = TCN_best_fold_info["auc"] / auc_sum

ensemble_probs_weighted = (
    w_mlp * mlp_probs +
    w_gru * gru_probs +
    w_tcn * tcn_probs
)
ensemble_pred_weighted = (ensemble_probs_weighted >= 0.5).astype(int)

acc_weighted = accuracy_score(y_test_tensor.numpy(), ensemble_pred_weighted)
auc_weighted = roc_auc_score(y_test_tensor.numpy(), ensemble_probs_weighted)

print(f"(2) [Weighted Voting Ensemble] Test Accuracy: {acc_weighted:.4f}, ROC AU

```

(1) [Soft Voting Ensemble] Test Accuracy: 0.9996, ROC AUC: 0.6822

(2) [Weighted Voting Ensemble] Test Accuracy: 0.9996, ROC AUC: 0.6821